

Assignment 1, Harmonic Analysis

Simon Gustafsson

22 September 2026

Problem 1

(a)

Proof.

$$\begin{aligned} M'_c(f)(x) &= \sup_{r>0} \frac{1}{r^d} \int_{x+(-r/2, r/2)^d} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{r^d} \int_{B(x, r\sqrt{d}/2)} |f(y)| dy && (x + (-r/2, r/2)^d \subseteq B(x, r\sqrt{d}/2)) \\ &= \sup_{\rho>0} \frac{d^{d/2}}{2^d \rho^d} \int_{B(x, \rho)} |f(y)| dy && (\rho = r\sqrt{d}/2) \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \sup_{\rho>0} \frac{1}{v_d \rho^d} \int_{B(x, \rho)} |f(y)| dy \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \cdot M_c(f)(x). \quad \square \end{aligned}$$

(b)

Proof.

$$\begin{aligned} M_c(f)(x) &= \sup_{r>0} \frac{1}{v_d r^d} \int_{B(x, r)} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{v_d r^d} \int_{x+(-r, r)^d} |f(y)| dy && (B(x, r) \subseteq x + (-r, r)^d) \\ &= \sup_{s>0} \frac{2^d}{v_d s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy && (s = 2r) \\ &= 2^d \cdot v_d^{-1} \sup_{s>0} \frac{1}{s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy \\ &= 2^d \cdot v_d^{-1} \cdot M'_c(f)(x). \quad \square \end{aligned}$$

Problem 2

(a)

Proof. Since clearly $g_q(x) := |f(x) - q| \in L^1_{\text{loc}}(\mathbb{R}^d)$, by Lebesgue Differentiation Theorem

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy = |f(x) - q| \quad \text{for a.e. } x \in \mathbb{R}^d. \quad (1)$$

Let A_q be the set where it fails, then $|A_q| = 0$ and the limit holds for all $x \in \mathbb{R}^d \setminus A_q$. $\square$

(b)

Proof. Let $A := \bigcup_{q \in \mathbb{Q}} A_q$, where each A_q is as in previous part. Since $\mathbb{Q}$ is countable, $|A| = 0$, and for every $x \in \mathbb{R}^d \setminus A$, (1) holds simultaneously for all $q \in \mathbb{Q}$. Fix $x \in \mathbb{R}^d \setminus A$ and $\varepsilon > 0$. Since f is real-valued and $\mathbb{Q}$ is dense in $\mathbb{R}$, choose $q \in \mathbb{Q}$ such that $|f(x) - q| < \varepsilon/2$.

$$\begin{aligned} & \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)| dy \\ & \leq \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} (|f(y) - q| + |f(x) - q|) dy \\ & = \limsup_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy + |f(x) - q| \quad (|f(x) - q| \text{ const. in } y) \\ & = |f(x) - q| + |f(x) - q| = 2|f(x) - q| < \varepsilon \end{aligned}$$

Since ε was arbitrary we are done. $\square$

(c)

Problem 3